Frontend Use Cases

Medicaid Claims Checker · test plan for the running app · status as of October 20, 2026

What this is. The agreed list of things to test in the running Medicaid Claims Checker through its web pages: the **Fetch Source** page (`http://localhost:4000/`, Manual Upload and Batch History) and the **Rules** page (`/rules`). Each use case names the test file to upload and what the app should show. Engine-only checks are out of scope.

59 use cases **40 Ready** — test data exists **1 Partial** — data made on demand **18 Gap** — no test file yet

Before you start

- The engine and Phoenix are running (`./start.sh`); the dev database has the 17 seeded active rules and the October 2026 NPPES import.
- **Upload copies, not the originals.** The app refuses a file or claim it has already received. Make a fresh set of copies for each round (see the companion guide *Repeat Testing with Mock Data*); paths below are relative to that round's folder.
- File sets: `mock_data/` (JSON claims, batches, rule files) and `x12_suite/files/` (X12/EDI/zip files). Exact expected results for every file are in each set's `manifest.json` and the `mock_data` README.
- The claims in `mock_data/claims/redundancy/` are written for the `redundancy_levels.dsl` rules, not the live rules; check those with the verify script.

J — JSON uploads

ID	Use case	Status	Test file(s)	What the app should show
J1	One claim in a .json file	**Ready**	`mock_data/claims/clean/office_visit.json`	"1 JSON claim(s) submitted for evaluation"; batch completes; claim is Clean.
J2	A list of claims `[ … ]`	**Gap**	`Test fixture only: json_upload/claims_list_with_bad_entries.json (also holds bad entries)`	Every claim in the list is evaluated.
J3	Claims wrapped as `{"claims": [ … ]}`	**Ready**	`mock_data/batches/mixed_batch.json, clean_batch.json`	Six (or four) claims evaluated; results as in the mock_data README.
J4	Claims mixed with non-claim entries	**Gap**	`Test fixture only: json_upload/claims_list_with_bad_entries.json`	Claims are evaluated; each non-claim entry is refused and named in the message.
J5	A JSON file that is not a claim	**Gap**	–	Refused: "no claim found (expected a claim_id or claim field)".
J6	An engine request (rulesText + claims)	**Gap**	`Test fixture only: json_upload/batch_redundancy_extended.json`	Refused as an engine request, not graded as a claim.
J7	Invalid JSON	**Gap**	–	Refused: "not valid JSON".
J8	Institutional / dental claim as JSON	**Ready**	`mock_data/claims/clean/inpatient_weekend_stay.json; single_rule/ExtremeAmounts.json`	Evaluated like a converted 837I; inpatient weekend stay is Clean.

X — X12 (.x12) uploads

ID	Use case	Status	Test file(s)	What the app should show
X1	837P professional claim	**Ready**	`x12_suite/files/single/01–37`	“1 X12/EDI claim(s) translated”; result per the X12 suite manifest.
X2	837I institutional claim	**Ready**	`x12_suite/files/formats/01_institutional_837i.x12, 06_…weekend.x12`	Translated and evaluated; 06 does not match WeekendOfficeVisit.
X3	837D dental claim	**Ready**	`x12_suite/files/formats/02_dental_837d.x12`	Translated and evaluated; Clean.
X4	Several claims in one transaction set	**Ready**	`x12_suite/files/multi/01_three_claims.x12, 03_…same_subscriber.x12`	Each claim becomes its own result row.
X5	Several transaction sets in one file	**Ready**	`x12_suite/files/multi/02_two_transaction_sets.x12`	Claims from both sets are evaluated.
X6	One-line file / Windows line endings	**Ready**	`x12_suite/files/formats/03_single_line_no_newlines.x12, 04_crlf_line_endings.x12`	Read exactly like a normal file; Clean.
X7	Broken envelope or file	**Ready**	`x12_suite/files/errors/01–03, 05, 06`	Refused with a clear error naming the problem; no claims evaluated.

E — EDI (.edi) uploads

ID	Use case	Status	Test file(s)	What the app should show
E1	A plain, clean .edi file	**Gap**	–	Treated exactly like .x12; Clean.
E2	Alternate delimiters (| and >)	**Ready**	`x12_suite/files/formats/05_alternate_delimiters.edi`	Translated; HighComplexityERVisit + UnauthorizedProcedure.
E3	A file that is not X12 at all	**Ready**	`x12_suite/files/errors/04_not_x12.edi`	Refused as not X12.

Z — Zip uploads

ID	Use case	Status	Test file(s)	What the app should show
Z1	Zip with .x12, .edi, .json and other files	**Ready**	`x12_suite/files/zip/01_mixed_bundle.zip`	The three claim files are evaluated; README.txt is ignored.
Z2	Zip with JSON files only	**Gap**	–	Every JSON claim inside is evaluated.
Z3	Path-traversal entry (../escape.x12)	**Ready**	`x12_suite/files/zip/02_path_traversal.zip`	Read as escape.x12; nothing is written to disk.
Z4	Zip with nothing usable / empty zip	**Gap**	–	Refused: no claims found.
Z5	Zip inside a zip	**Gap**	–	The inner zip is ignored.
Z6	Corrupt zip	**Gap**	–	Refused with a clear error.
Z7	More than 15,000 claims	**Partial**	`Generated on demand: cd phoenix_web; elixir test/fixtures/x12_suite/generate.exs --large 15000`	Up to 15,000 claims are evaluated; a larger batch is refused (limit 15,000).

U — Upload behaviour

ID	Use case	Status	Test file(s)	What the app should show
U1	Several files of mixed types at once	**Ready**	`Pick files from several sets`	Each file's claims are evaluated in one batch.
U2	Unsupported file type (.txt, .pdf)	**Gap**	`Any non-claim file`	"File type not accepted. Allowed: .json, .x12, .edi, .zip".
U3	File over 10 MB	**Gap**	`Create on demand; do not commit`	"File is too large (max 10 MB)."
U4	Same file uploaded twice	**Ready**	`Any file, uploaded twice`	Second time: "Nothing was evaluated: every claim was already submitted…", a red notice in Batch History, and an email to active admin users (real inboxes when Brevo mail is configured; otherwise /dev/mailbox).
U5	Same claim in a different file	**Gap**	–	That claim is refused, naming the earlier batch; other claims are evaluated.
U6	Same claim twice in one file	**Gap**	–	The second copy is refused ("appears twice in this submission").
U7	Void / replacement (frequency 8 / 7)	**Gap**	–	Accepted: a void or replacement is not a duplicate of the original.
U8	Re-uploading test data as new claims	**Ready**	`retag_test_data.exs (see the Repeat Testing guide)`	Tagged copies are accepted as new claims.

R — Business rules (17 active rules)

ID	Use case	Status	Test file(s)	What the app should show
R1	Clean claim	**Ready**	`mock_data/claims/clean/*; x12_suite/files/single/01, 02`	No rule matches; claim is Clean.
R2	Each rule on its own	**Ready**	`mock_data/claims/single_rule/<RuleName>.json (one per rule)`	Exactly the named rule matches.
R3	Several rules on one claim	**Ready**	`mock_data/claims/multi_rule/*; x12_suite/files/single/27_many_rules.x12`	Every expected rule matches.
R4	Problem on the 2nd service line	**Ready**	`x12_suite/files/single/35, 36, 37`	Rules check every line, not just the first.
R5	Exact thresholds	**Ready**	`x12_suite/files/single/28, 29, 30, 34`	"Greater than" limits do not match at exactly the limit.
R6	Weekend office visit variants	**Ready**	`x12_suite/files/single/24, 31, 32, 33`	Saturday/Sunday office visits match; weekday or non-office do not.
R7	Weekend inpatient stay	**Ready**	`x12_suite/files/formats/06; mock_data/claims/clean/inpatient_weekend_stay.json`	WeekendOfficeVisit does not match.
R8	New-provider rules (tenure)	**Ready**	`mock_data/claims/single_rule/NewProviderLargeSubmission.json, TaxonomyMismatchNewProvider.json`	Match on JSON claims (tenure comes only from JSON).
R9	Rule that cannot be decided	**Ready**	`x12_suite/files/single/21_missing_taxonomy_new_provider.x12`	Shown as "Not checked", naming the missing field.

R10	Patient age out of range / missing DOB	**Ready**	`x12_suite/files/single/11, 12, 13`	MissingRequiredFields or PatientAgeOutOfRange.
R11	Void / replacement against business rules	**Gap**	–	To be defined: how rules treat frequency 7/8 claims.

N — NPPES provider checks

ID	Use case	Status	Test file(s)	What the app should show
N1	Active provider	**Ready**	`Any clean claim`	NPPES check passes.
N2	Malformed NPI (length / checksum)	**Ready**	`x12_suite/files/single/14, 15`	Rules flag the NPI; NPPES rejects it.
N3	NPI never issued	**Ready**	`x12_suite/files/single/16; mock_data/claims/nppes/unissued_npi.json`	“not found in NPPES”.
N4	Deactivated provider	**Ready**	`x12_suite/files/single/17; mock_data/claims/nppes/deactivated_provider.json`	“was deactivated”.
N5	Service before the deactivation	**Ready**	`x12_suite/files/single/18`	Accepted.
N6	Deactivated part-way through a claim	**Ready**	`x12_suite/files/single/18b`	Rejects the later line’s date.
N7	Reactivated provider / service in the gap	**Ready**	`x12_suite/files/single/19, 19b; mock_data/claims/nppes/reactivated_provider.json, reactivation_gap.json`	Accepted after reactivation; rejected inside the gap.
N8	Service before the NPI was issued	**Ready**	`x12_suite/files/single/09; mock_data/claims/nppes/before_enumeration.json`	“was not issued until …”.
N9	Provider dropped from the NPPES file	**Ready**	`mock_data/claims/nppes/dropped_provider.json`	“has not been in the NPPES file since …” (needs the Oct 2026 import).
N10	No billing NPI	**Ready**	`x12_suite/files/single/25`	MissingRequiredFields.
N11	Deactivated rendering provider	**Ready**	`x12_suite/files/single/26`	NPPES rejects the rendering provider.

P — Rules page

ID	Use case	Status	Test file(s)	What the app should show
P1	Feature examples	**Ready**	`mock_data/rules/examples.dsl`	Each rule parses, passes the schema check and saves.
P2	Redundancy levels	**Ready**	`mock_data/rules/redundancy_levels.dsl (add in file order)`	Exact duplicate, subsumption, shadowed and overlap are each reported where the file’s comments say.
P3	Rule with a syntax error	**Gap**	–	Refused with the error position.
P4	Rule naming an unknown field	**Gap**	–	Kept as a draft, with a suggested field name.

Gaps to fill

No uploadable test file exists yet for: J2, J4–J7 (JSON edge cases; three exist only as Phoenix test fixtures), E1, Z2, Z4–Z6 (zip edge cases), U2, U3, U5–U7 (upload behaviour; U5 and U7 are covered by automated Phoenix tests), R11 (a rule decision is needed first), P3, P4 (Rules page errors). Z7 is generated on demand rather than stored, because of its size.